Failure of exterior observability at equal bulk and surface wave speeds in an annulus

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Abstract

We consider a wave equation in a planar annulus, with a Dirichlet condition on the outer circle and a dynamic wave equation on the inner circle. The bulk and surface propagation speeds are both one. For every finite observation time, we construct smooth real solutions with unit initial energy whose exterior normal-derivative observation tends to zero. The construction uses exact separated solutions with angular order m and frequency $\lambda_m = m + O(m^{-1/3})$. Large-order Bessel asymptotics at the inner turning point give a lower bound for the surface energy, while an exact Wronskian identity fixes the outer flux. Thus exterior observability fails at equality of the two speeds in this annulus.

1 The model and the result

Dynamic boundary conditions describe a vibrating surface coupled to a wave in the surrounding medium. We ask whether measurements on the exterior boundary determine the entire bulk and surface energy. Let

$$\Omega = \{x \in \mathbb{R}^2 : 1 < |x| < 2\}, \qquad \Gamma_1 = \{|x| = 1\}, \qquad \Gamma_0 = \{|x| = 2\}.$$

The equal-speed system is

$$\begin{aligned} z_{tt} - \Delta z &= 0 && \text{in } \Omega, \\ z &= 0 && \text{on } \Gamma_0, \\ z_\Gamma &= z|_{\Gamma_1}, \\ z_{\Gamma,tt} - \Delta_\Gamma z_\Gamma + \partial_\nu z &= 0 && \text{on } \Gamma_1. \end{aligned} \tag{1}$$

Here ν is the outward normal of the annulus and Δ_Γ is the Laplace operator of the unit circle. In polar coordinates, $\partial_\nu = -\partial_r$ on Γ_1, $\partial_\nu = \partial_r$ on Γ_0, and $\Delta_\Gamma = \partial_{\theta\theta}$. The energy and observation are

$$E_z(t) = \int_\Omega \left(|z_t|^2 + |\nabla z|^2\right) dx + \int_{\Gamma_1} \left(|z_{\Gamma,t}|^2 + |\nabla_\Gamma z_\Gamma|^2\right) dS, \tag{2}$$

$$\mathcal{O}_T(z) = \int_0^T \int_{\Gamma_0} |\partial_\nu z|^2 \, dS \, dt. \tag{3}$$

All solutions constructed below are real and smooth up to both boundaries and for all real times.

Baudouin, Dardé, Ervedoza and Mercado study the annular system with surface mass α and tangential coefficient β [1, Eq. (1.3), pp. 60–61]. Their Theorem 2.2 treats $\alpha < \beta$, while Theorem 2.4 gives failure of observability for $\alpha > \beta$. Neither theorem includes $\alpha = \beta = 1$, which is (1). The review of Chorfi, Maniar and Morales [2, Sec. 5.1] explicitly leaves the equal-speed case open. The result here answers the concrete annular question in [4] negatively.

Theorem 1. *There is an infinite set $\mathcal{M}$ of positive integers and smooth real solutions w_m of (1), $m \in \mathcal{M}$, such that*

$$E_{w_m}(0) = 1, \qquad \mathcal{O}_T(w_m) \le CTm^{-1/3} \quad \text{for every finite } T > 0,$$

where C is independent of m and T. Consequently, for no finite $T > 0$ does a finite constant C_T satisfy $E_z(0) \le C_T \mathcal{O}_T(z)$ for all smooth compatible solutions. In particular, no finite threshold time T_0 has this property for every $T > T_0$.

We construct exact modes rather than approximate solutions. Their radial equation has a turning point near the inner circle when λ is close to m. The inner boundary equation can be solved with a frequency correction of order $m^{-1/3}$. On a subsequence, the resulting surface kinetic energy grows while the unnormalized exterior flux remains fixed.

2 An exact family of radial modes

Let m be a positive integer and J_m, Y_m the conventional Bessel functions of order m. For $\lambda > 0$, the radial equation for angular dependence $\cos(m\theta)$ is

$$f'' + r^{-1} f' + (\lambda^2 - m^2/r^2) f = 0. \tag{4}$$

To impose the outer Dirichlet condition, define

$$F_m(r;\lambda) = J_m(\lambda r) Y_m(2\lambda) - Y_m(\lambda r) J_m(2\lambda), \qquad 1 \le r \le 2. \tag{5}$$

The Bessel equation gives (4) and $F_m(2;\lambda) = 0$. The Wronskian identity [3, Eq. 10.5.2]

$$J_m(x) Y_m'(x) - J_m'(x) Y_m(x) = \frac{2}{\pi x}$$

also gives the exact radial derivative

$$\partial_r F_m(2;\lambda) = \lambda\big[J_m'(2\lambda) Y_m(2\lambda) - Y_m'(2\lambda) J_m(2\lambda)\big] = -\frac{1}{\pi}. \tag{6}$$

In particular F_m is nonzero. Bessel primes mean argument derivatives; radial derivatives are written as $\partial_r F_m$.

The inner dynamic condition for a mode $\sin(\lambda t) F_m(r;\lambda)\cos(m\theta)$ is exactly

$$G_m(\lambda) := \partial_r F_m(1;\lambda) + (\lambda^2 - m^2) F_m(1;\lambda) = 0. \tag{7}$$

Indeed the inner outward normal is $-\partial_r$, so its boundary equation is $(m^2 - \lambda^2) F_m(1;\lambda) - \partial_r F_m(1;\lambda) = 0$. It remains to find roots near $\lambda = m$ with a sufficiently large inner value.

Lemma 1. *Set*

$$\kappa = \sqrt{3} - \frac{\pi}{3}, \quad \phi_m = m\kappa - \frac{\pi}{4}, \quad A = \left(\frac{2}{\pi\sqrt{3}}\right)^{1/2},$$

Let Ai, Bi *be the conventional Airy functions, and put*

$$a = 2^{1/3}\,\mathrm{Ai}(0) > 0, \qquad b = 2^{1/3}\,\mathrm{Bi}(0) > 0,$$
$$p = -2^{2/3}\,\mathrm{Ai}'(0), \qquad q = 2^{2/3}\,\mathrm{Bi}'(0).$$

For every fixed $M > 0$, as positive integers m tend to infinity, the following estimates hold uniformly for $\lambda = m + sm^{-1/3}$, $|s| \le M$:

$$J_m(\lambda) = am^{-1/3} + o(m^{-1/3}), \qquad Y_m(\lambda) = -bm^{-1/3} + o(m^{-1/3}), \tag{8}$$
$$J_m'(\lambda) = pm^{-2/3} + o(m^{-2/3}), \qquad Y_m'(\lambda) = qm^{-2/3} + o(m^{-2/3}), \tag{9}$$
$$J_m(2\lambda) = Am^{-1/2}(\cos\phi_m + o(1)), \qquad Y_m(2\lambda) = Am^{-1/2}(\sin\phi_m + o(1)). \tag{10}$$

Proof. We first use standard expansions only at the fixed arguments m and $2m$, and then prove the required uniformity from the Bessel equation. In the uniform Airy expansions [3, Eqs. 10.20.4, 10.20.5, 10.20.7, 10.20.8], the Bessel argument is $x = m\eta$, with normalized argument $\eta > 0$. The value formulas and their separately stated argument-derivative formulas, evaluated at $\eta = 1$, give (8) and (9) at $\lambda = m$. To check the prefactors, the defining turning variable satisfies $\zeta(\eta) = 2^{1/3}(1-\eta)+O((1-\eta)^2)$; hence the continuous limits of $(4\zeta/(1-\eta^2))^{1/4}$ and $(2/\eta)((1-\eta^2)/(4\zeta))^{1/4}$ are $2^{1/3}$ and $2^{2/3}$. The leading coefficient functions in the value and argument-derivative expansions equal one, and the remaining coefficients are smooth at $\zeta = 0$ [3, Eqs. 10.20.10–10.20.14]. The Airy values and signs are given in [3, Eqs. 9.2.3–9.2.6].

The fixed-angle Debye expansions [3, Eqs. 10.19.5–10.19.7] use the argument $x = m\sec\beta$ and phase $m(\tan\beta - \beta) - \pi/4$, where the angle $\beta \in (0, \pi/2)$ is fixed. At $\beta = \pi/3$, the argument is $2m$ and the phase is ϕ_m, so they give

$$\begin{aligned} J_m(2m) &= Am^{-1/2}\cos\phi_m + O(m^{-3/2}), \\ Y_m(2m) &= Am^{-1/2}\sin\phi_m + O(m^{-3/2}), \\ |J_m'(2m)| + |Y_m'(2m)| &= O(m^{-1/2}). \end{aligned}$$

These error estimates are absolute, including at zeros of the leading trigonometric factors.

For either $y = J_m$ or $y = Y_m$, its equation is

$$y''(x) = -x^{-1}y'(x) - (1 - m^2/x^2)y(x).$$

Put $h = Mm^{-1/3}$. On $|x - m| \le h$, for all sufficiently large m,

$$x^{-1} \le 2/m, \qquad |1 - m^2/x^2| \le C_M m^{-4/3}.$$

Let $U = \sup|y|$ and $V = \sup|y'|$ on this interval. Integrating from m, on either side, gives

$$U \le |y(m)| + hV, \qquad V \le |y'(m)| + h(2V/m + C_M m^{-4/3}U).$$

Substitute the first bound into the second. The coefficient of V on the right tends to zero and can be absorbed. The endpoint scales give $V = O_M(m^{-2/3})$ and $U = O_M(m^{-1/3})$. Integrating once more yields

$$y(x) - y(m) = O_M(m^{-1}), \qquad y'(x) - y'(m) = O_M(m^{-2}).$$

These changes are smaller than the respective leading scales.

On $|x - 2m| \le 2h$, both equation coefficients are bounded. If $W = \sup(|y| + |y'|)$ there, integration gives

$$W \le |y(2m)| + |y'(2m)| + C(2h)W.$$

Since $h \to 0$, absorption gives $W = O_M(m^{-1/2})$, and $y(x) - y(2m) = O_M(m^{-5/6}) = o(m^{-1/2})$. This proves all the uniform shifted estimates without differentiating an asymptotic remainder. □

Lemma 2. *There exist constants $M, \delta > 0$, an infinite set $\mathcal{M}$ of positive integers, and exact roots $\lambda_m > 0$ of (7) such that, for all sufficiently large $m \in \mathcal{M}$,*

$$|\lambda_m - m| \le Mm^{-1/3}, \qquad |F_m(1;\lambda_m)| \ge \frac{A\delta}{2}m^{-5/6}.$$

Proof. Define the two phase factors

$$S_m = a\sin\phi_m + b\cos\phi_m, \qquad B_m = p\sin\phi_m - q\cos\phi_m.$$

At the inner circle, the determinant (5) and its radial derivative read

$$F_m(1;\lambda) = J_m(\lambda)Y_m(2\lambda) - Y_m(\lambda)J_m(2\lambda),$$
$$\partial_r F_m(1;\lambda) = \lambda\big[J'_m(\lambda)Y_m(2\lambda) - Y'_m(\lambda)J_m(2\lambda)\big].$$

Lemma 1 therefore gives, uniformly on each fixed $|s| \le M$,

$$F_m(1; m + sm^{-1/3}) = Am^{-5/6}(S_m + o(1)), \tag{11}$$
$$\partial_r F_m(1; m + sm^{-1/3}) = Am^{-1/6}(B_m + o(1)). \tag{12}$$

The second formula includes the factor $\lambda/m = 1 + o(1)$ from radial differentiation.

We choose a subsequence where the inner value does not cancel. Write $S_m = R\sin(\phi_m + \vartheta)$, where $R = (a^2 + b^2)^{1/2} > 0$. Since $0 < \kappa < \pi$,

$$\sin\kappa = \sin(x+\kappa)\cos x - \cos(x+\kappa)\sin x$$

implies $|\sin x| + |\sin(x+\kappa)| \ge \sin\kappa > 0$. Thus in each disjoint pair $m = 2k, 2k+1$, $k \ge 1$, at least one index satisfies

$$|S_m| \ge \delta := \frac{R\sin\kappa}{2}.$$

Select one such index from each pair to obtain the infinite set $\mathcal{M}$. This argument needs no equidistribution of the phases.

The exact expansion $\lambda^2 - m^2 = 2sm^{2/3} + s^2m^{-2/3}$, together with (11)–(12), gives

$$m^{1/6}G_m(m + sm^{-1/3}) = A(B_m + 2sS_m + o(1)) \tag{13}$$

uniformly for each fixed $|s| \le M$. Choose M once and for all with

$$2M\delta > |p| + |q| + 2.$$

For sufficiently large $m \in \mathcal{M}$, the right side of (13) has opposite signs at $s = -M$ and $s = M$: the terms $\mp 2MS_m$ dominate B_m and the uniform remainder. Both frequency endpoints are positive. The intermediate value theorem therefore supplies an exact root λ_m in that interval. Finally, (11) and $|S_m| \ge \delta$ give the asserted inner-value bound throughout the same interval, in particular at its root. □

3 Energy and exterior observation

Proof of Theorem 1. Discard the finitely many low orders excluded in Lemma 2, retaining the notation $\mathcal{M}$. Put

$$f_m(r) = F_m(r;\lambda_m), \qquad z_m(t,r,\theta) = \sin(\lambda_m t) f_m(r)\cos(m\theta).$$

The polar Laplacian and (4) give the bulk wave equation. The outer trace is zero, and (7) gives the inner equation with the stated outward normal. Integer angular order ensures periodicity. Every Bessel argument is positive, so the modes are real and smooth on the closed annulus for all real times. These exact identities also give every initial compatibility condition.

At $t = 0$, displacement is zero and velocity is $\lambda_m f_m(r)\cos(m\theta)$. Since $\int_0^{2\pi}\cos^2(m\theta)\,d\theta = \pi$, the full energy (2) is

$$E_{z_m}(0) = \pi\lambda_m^2\left(\int_1^2 |f_m(r)|^2 r\,dr + |f_m(1)|^2\right). \tag{14}$$

For all sufficiently large selected m, $\lambda_m \geq m/2$. The surface term alone and Lemma 2 yield

$$E_{z_m}(0) \geq \frac{\pi A^2 \delta^2}{16} m^{1/3} > 0. \tag{15}$$

On the outer circle $dS = 2\,d\theta$, and (6) gives

$$\mathcal{O}_T(z_m) = \frac{2}{\pi} \int_0^T \sin^2(\lambda_m t)\, dt \leq \frac{2T}{\pi}. \tag{16}$$

Normalize by $w_m = z_m / \sqrt{E_{z_m}(0)}$. Energy and observation are quadratic, so (15)–(16) give

$$E_{w_m}(0) = 1, \qquad \mathcal{O}_T(w_m) \leq \frac{32T}{\pi^2 A^2 \delta^2} m^{-1/3} \longrightarrow 0$$

for every fixed finite $T > 0$. A finite C_T in the proposed inequality would require $1 \leq C_T \mathcal{O}_T(w_m)$ for this sequence, which is impossible. Choosing any finite positive $T > T_0$ also excludes every finite threshold T_0. □

The conclusion concerns precisely the unit-speed system on the annulus with radii one and two. The estimate above establishes failure of energy observability; it does not identify an optimal weaker observation norm or a stabilization rate.

AI-use statement. OpenAI Codex assisted with literature retrieval, the spectral construction, checking the argument, and drafting this manuscript. The conclusion is supported by the analytic proof above; no numerical calculation is used as a proof premise. Independent mathematical review is requested.

References

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[3] NIST Digital Library of Mathematical Functions, Sections 9.2, 10.2, 10.5, 10.19, 10.20 and 10.41. https://dlmf.nist.gov/. Consulted 12 October 2026.

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